

Name _____ Date _____ Period _____

Parent Functions I: $\sqrt{x}$ and $|x|$ - Practice

Algebra II | Unit 1 | Lesson 4 Practice | TEK A2.2.A

Objective: I can evaluate and compare the square root and absolute value parent functions, and reason about their solution counts.

Section 1 — Recall

1. Evaluate: $\sqrt{36} =$ _____ $\sqrt{0} =$ _____ $|-9| =$ _____ $|0| =$ _____
2. Complete: $f(x) = \sqrt{x}$ at $x = 1, 4, 16$.
 $f(1) =$ _____ $f(4) =$ _____ $f(16) =$ _____

Section 2 — Apply

3. Name every x where $|x| = 10$. How many are there? Compare to every x where $\sqrt{x} = 10$ - how many?
 $|x| = 10$: _____ $\sqrt{x} = 10$: _____
4. On $[4, 16]$, give the minimum and maximum of $f(x) = \sqrt{x}$.
min: _____ max: _____

Section 3 — Challenge

5. Without graphing, explain how many solutions each equation has, and why: $|x - 3| = 7$, and $\sqrt{x - 3} = 7$.

 $|x - 3| = 7$: _____ $\sqrt{x - 3} = 7$: _____
6. A pool's depth follows $d(x) = \sqrt{x}$ for x from 0 to 9 (x = distance from shore in yards, d = depth in feet). At what distance is the pool exactly 3 ft deep?

Distance: _____

Section 4 — Explain It

7. A classmate says: " $\sqrt{x^2}$ is the same as x for every real number x , since squaring and square-rooting undo each other." Is the classmate correct? Test $x = -5$ and explain.
